FRONT WIPERS/WASHERS (Continued)

of the two wiper pivot shafts, which extend through the cowl plenum cover/grille panel located near the base of the windshield.

- **Front Wiper Blade** - The two front wiper blades are secured to the two front wiper arms with integral latches, and are parked on the glass near the bottom of the windshield when the front wiper system is not in operation.
- **Front Wiper Module** - The front wiper pivot shafts are the only visible components of the front wiper module. The remainder of the module is concealed within the cowl plenum area beneath the cowl plenum cover/grille panel. The front wiper module includes the wiper module bracket, three rubber-isolated wiper module mounts, the front wiper motor, the front wiper motor crank arm, the front wiper drive link, the front wiper connector link, and the two front wiper pivots.
- **Right Multi-Function Switch** - The right multi-function switch is located near the top of the steering column, just below the steering wheel. The right multi-function switch includes a control stalk with a control knob on the end that extends through a dedicated opening on the right side of the steering column shrouds. The right multi-function switch is dedicated to providing all of the driver controls for the front wiper and washer systems.
- **Washer Reservoir** - The molded plastic washer reservoir is secured by two screws through integral mounting tabs to the top of the left front fender wheel house, in the left front corner of the engine compartment. The washer reservoir filler neck is accessed from the left front corner of the engine compartment.

Hard wired circuitry connects the front wiper and washer system components to the electrical system of the vehicle. These hard wired circuits are integral to several wire harnesses, which are routed throughout the vehicle and retained by many different methods. These circuits may be connected to each other, to the vehicle electrical system and to the front wiper and washer system components through the use of a combination of soldered splices, splice block connectors, and many different types of wire harness terminal connectors and insulators. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

OPERATING MODES

The components of the wiper and washer system are designed to work in concert to provide the following operating modes:

- **Continuous Wipe Modes** - The control stalk of the right multi-function switch has two continuous wipe positions, Low and High. When selected, these switch positions will cause the two-speed wiper motor to operate in a continuous low or high speed cycle.
- **Intermittent Wipe Mode** - The control stalk of the right multi-function switch has an intermittent wipe position, and the control knob has several minor detent positions which will each cause the wiper system to operate at different delay intervals between complete wipe cycles. The internal circuitry of the right multi-function switch provides the intermittent wipe mode and adjustable delay intervals between wipe cycles of about one second to about fifteen seconds.
- **Mist Wipe Mode** - The control stalk of the right multi-function switch has a momentary mist wipe position that will operate the front wipers for a single complete cycle, then park the wiper blades near the base of the windshield.
- **Washer Mode** - When the control stalk of right multi-function switch is pulled toward the steering wheel to the momentary Wash position while the front wiper system is operating, washer fluid will be dispensed onto the windshield glass through the washer nozzles for as long as the front washer pump is energized.
- **Wipe-After-Wash Mode** - When the control stalk of right multi-function switch is pulled toward the steering wheel to the momentary Wash position while the front wiper system is not operating, the internal circuitry of the right multi-function switch provides a wipe-after-wash feature which will operate the front washer pump/motor and the front wipers for as long as the washer system is activated, then provide one or two additional wipe cycles after the washer system is deactivated before parking the front wiper blades near the base of the windshield.

OPERATION

The front wiper and washer system is designed to provide the vehicle operator with a convenient, safe, and reliable means of maintaining visibility through the windshield glass. The various components of this system are designed to convert electrical energy produced by the vehicle electrical system into the mechanical action of the wiper blades to wipe the outside surface of the glass, as well as into the hydraulic action of the washer system to apply washer fluid stored in an on-board reservoir to the area of the glass to be wiped. When combined, these components provide the means to effectively maintain clear visibility for the vehicle operator by removing excess accumulations of rain, snow, bugs, mud, or other minor debris from the outside windshield glass surface that might be encountered while driving the